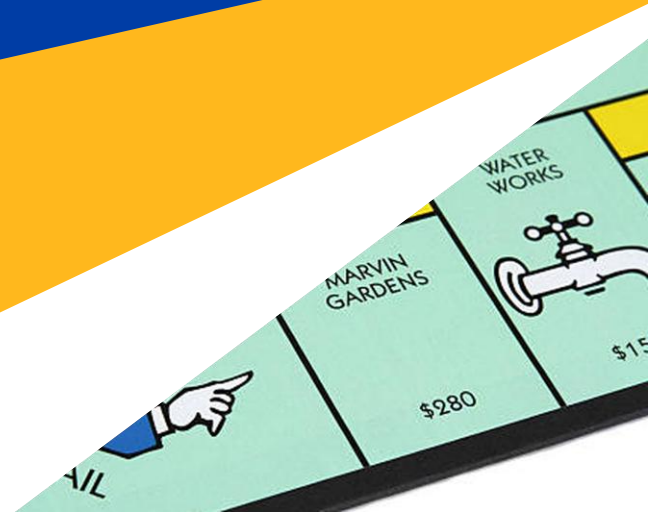
Chapter 12

Monopoly

ECON 003
UC Riverside · Summer 2026



Where We Left Off

Chapter 7 assumed every seller was a PRICE TAKER. Today we remove that one condition and nothing else.

WHAT CHAPTER 7 PROVED

In PERFECT COMPETITION, P = MR = MC, firms earn ZERO ECONOMIC PROFIT in the long run, and SOCIAL SURPLUS is maximized.

THE ASSUMPTION WE DROP

Many sellers, none big enough to move the price. A MONOPOLY is one seller of a good with no close substitutes — and it is therefore a PRICE MAKER.

WHAT BREAKS

The monopolist maximizes profit exactly as the competitive firm did — and produces LESS at a HIGHER price, with DEADWEIGHT LOSS.

MARKET POWER — the ability to set the price. It is the only thing that changes today, and it is enough to break the result.

One Seller, Six Moves

A MONOPOLIST picks a QUANTITY and lets DEMAND pick the price. Everything today follows from that.



Price taker vs price maker



Where market power comes from



Why MR is below price



Choosing Q, then reading P



What it costs, and what policy can do

Sections 12.1 → 12.5 and 12.7: MARKET POWER · BARRIERS TO ENTRY · MARGINAL REVENUE · MR = MC · DEADWEIGHT LOSS · REGULATION

The competitive firm chose only a QUANTITY. So does the monopolist — the difference is that its choice moves the PRICE.

PART 1

One Seller and Market Power

Two Market Structures, Side by Side

MONOPOLY — one seller of a good or service with NO CLOSE SUBSTITUTES.

	Perfect competition	Monopoly
Number of sellers	Many	One
The product	Identical across sellers	No close substitutes
Barriers to entry	None — free entry and exit	High
Price	PRICE TAKER: P = MR = MC	PRICE MAKER: P > MR = MC
Demand facing the firm	Horizontal — perfectly elastic	The whole market curve, downward-sloping
Long-run profit	Zero economic profit	Can stay positive indefinitely
Social surplus	Maximized	Not maximized

Read the PRICE row: the competitive firm is HANDED P = MR = MC. The monopolist SETS P above MR = MC. Every other row follows.

Four Reasons Nobody Else Can Enter

BARRIERS TO ENTRY — circumstances that prevent potential competitors from entering the market.

LEGAL

The state grants exclusivity. A PATENT for an invention, a COPYRIGHT for a work, a license for a route or a frequency.

KEY RESOURCE

Own the input and you own the output. Alcoa and bauxite; De Beers and rough diamonds.

ECONOMIES OF SCALE

ATC still falling at the size of the whole market. A NATURAL MONOPOLY — the next page.

NETWORK EXTERNALITIES

The product is worth more because others use it. A rival with a better product still starts with nobody on it.

Chapter 7's ENTRY drove economic profit to zero. A BARRIER TO ENTRY is precisely what stops that mechanism running.

A Barrier With a Price Tag On It



One firm bought one old drug and raised the price from $13.50 a tablet to $750. Why could it?

Nothing about the drug changed. Daraprim came off patent decades ago, so anyone may make it.

The BARRIER TO ENTRY was regulatory delay: a new manufacturer must take a generic through FDA approval first, and that can take more than two years.

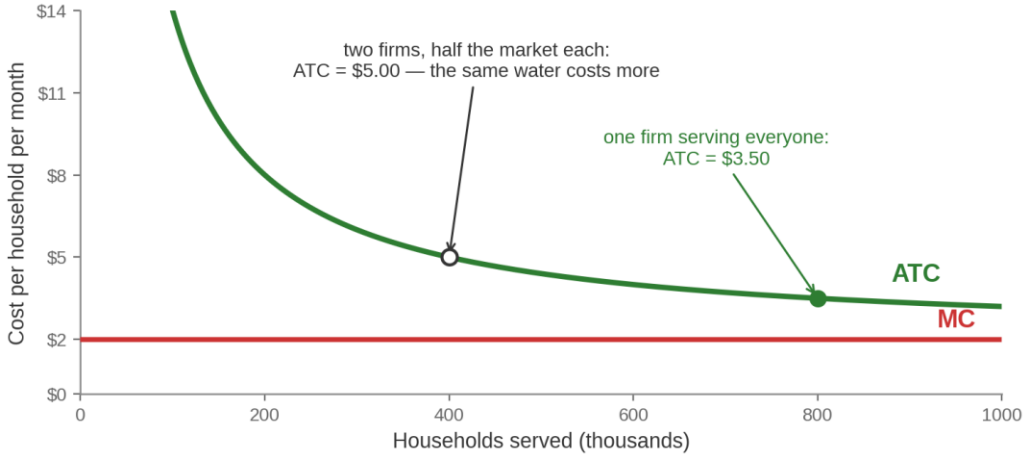
For those two years the buyer had no close substitute — which is the definition of a MONOPOLY, arrived at without a patent, a key resource or a network.

A barrier does not have to be permanent to confer MARKET POWER. It only has to last longer than the buyer can wait.

Image: Acemoglu, Laibson & List, Microeconomics 3e

When One Firm Really Is Cheaper Than Two

A NATURAL MONOPOLY has ECONOMIES OF SCALE over the whole market: ATC is still falling when one firm has served everybody.



Split the market and ATC rises $3.50 → $5.00. The BARRIER TO ENTRY here is the COST CURVE itself — no rule is needed.

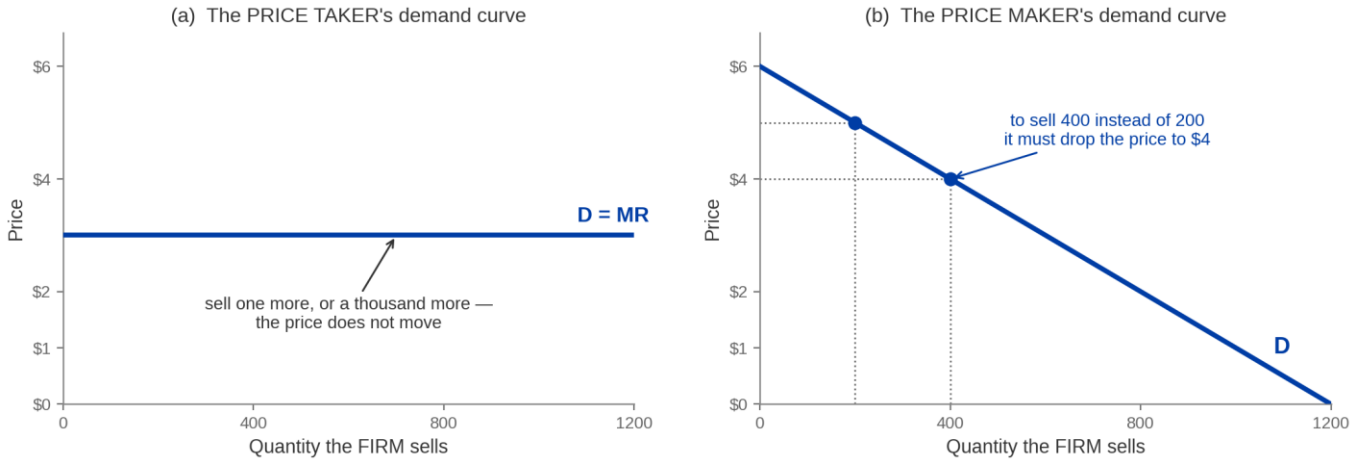
Worked example. Fixed cost $1.2m a month, marginal cost $2 a household

PART 2

The Monopolist’s Problem

The Curve the Firm Faces Is the Whole Market

A PRICE TAKER faces a horizontal demand curve, so D = MR. A MONOPOLIST faces the MARKET demand curve, so it slopes DOWN.



The MONOPOLIST IS THE INDUSTRY — it cannot sell more without cutting the price, and it cannot cut the price for only the last buyer.

Revenue and Cost at Every Output

MARGINAL REVENUE is the change in TOTAL REVENUE from one more unit — and it is NOT the price.

Q (m pills)	Price	Total revenue	Marginal revenue	Total cost	Marginal cost	ATC
100	$5.50	$550	$5.00	$110	$1.00	$1.100
200	$5.00	$1,000	$4.00	$210	$1.00	$1.050
300	$4.50	$1,350	$3.00	$310	$1.00	$1.033
400	$4.00	$1,600	$2.00	$410	$1.00	$1.025
500	$3.50	$1,750	$1.00	$510	$1.00	$1.020
600	$3.00	$1,800	$0.00	$610	$1.00	$1.017
700	$2.50	$1,750	−$1.00	$710	$1.00	$1.014

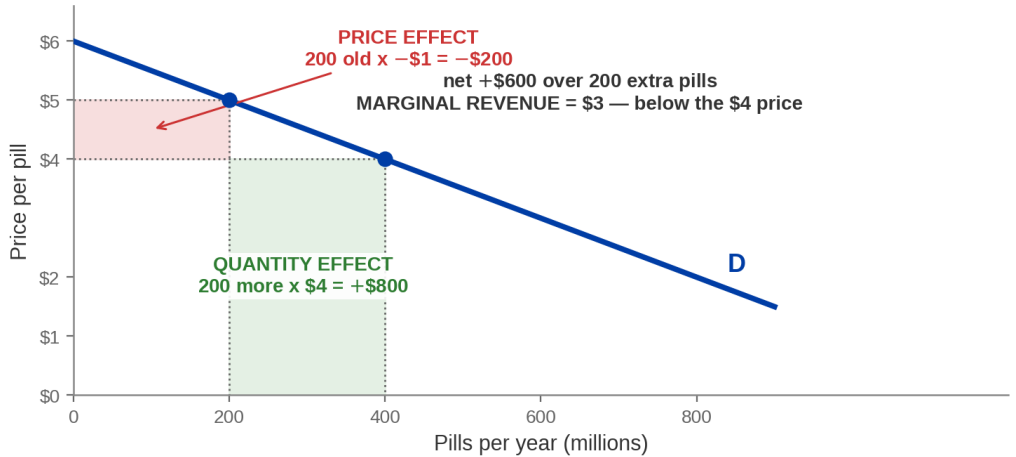
At every row MARGINAL REVENUE is BELOW price, and by Q = 600 it is zero: selling more would REDUCE total revenue.

Cut the Price and Two Things Happen at Once



Cut the Price and Two Things Happen at Once

QUANTITY EFFECT: more units sold. PRICE EFFECT: every unit you were already selling now earns less.

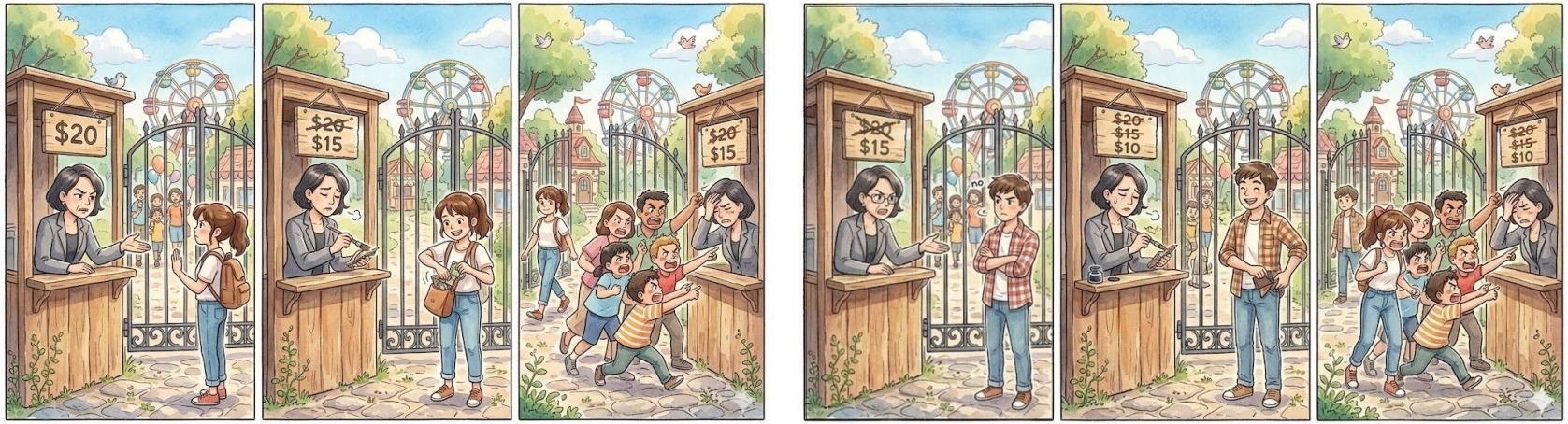


MARGINAL REVENUE = ($800 − $200) ÷ 200 = $3 a pill — below the $4 price, because the PRICE EFFECT is subtracted from it.

Worked on the book's own $5 → $4 example, Acemoglu, Laibson & List, ch. 12

Why MR Falls Twice as Fast as the Price (Not Required)

SECOND-ROUND OUTPUT EFFECT: the new buyer pays $5 LESS than the last one did.
SECOND-ROUND PRICE EFFECT: the refund now goes to ONE MORE earlier buyer.

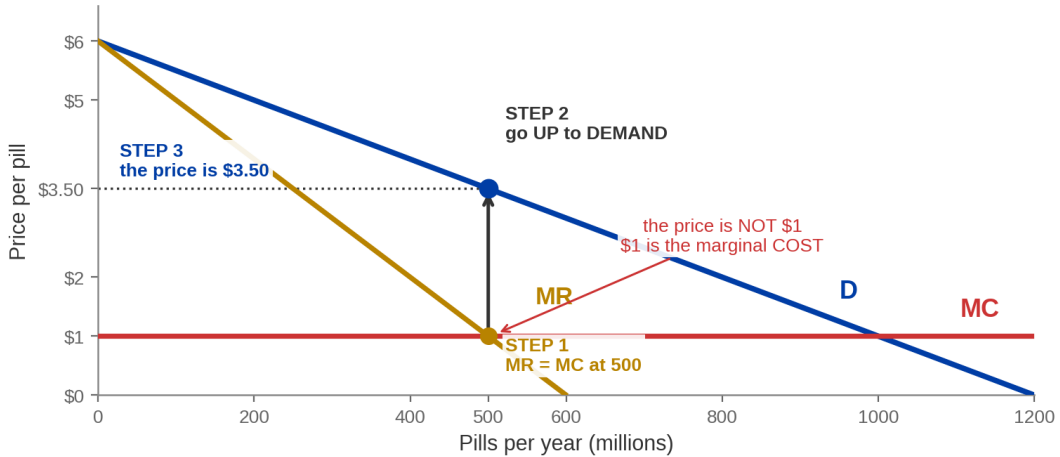


Every $5 cut costs $5 MORE in refunds and brings in $5 LESS at the gate — so MARGINAL REVENUE falls $10 each time the PRICE falls $5.

Our own illustration — the book states the result and does not derive it

Three Steps

STEP 1 find Q where MR = MC. STEP 2 go UP to the DEMAND curve. STEP 3 read the PRICE off the vertical axis.

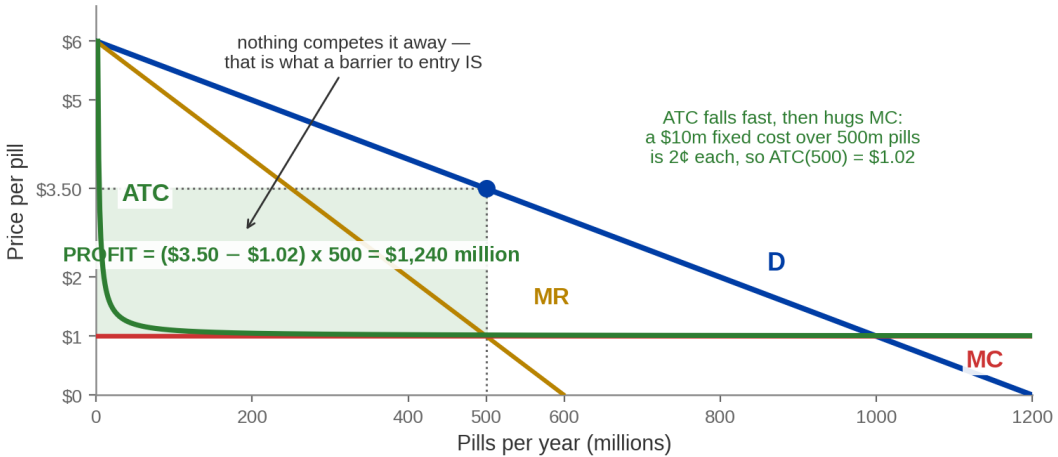


MR = MC at 500 million pills, and DEMAND sets the price at $3.50. The rule is P > MR = MC.

Market drawn to Acemoglu, Laibson & List, Microeconomics 3e, ch. 12

Profit Is a Rectangle, and Nothing Erodes It

PROFIT = (P − ATC) × Q, exactly as in Chapter 7. What is new is that no ENTRY arrives to compete it away.

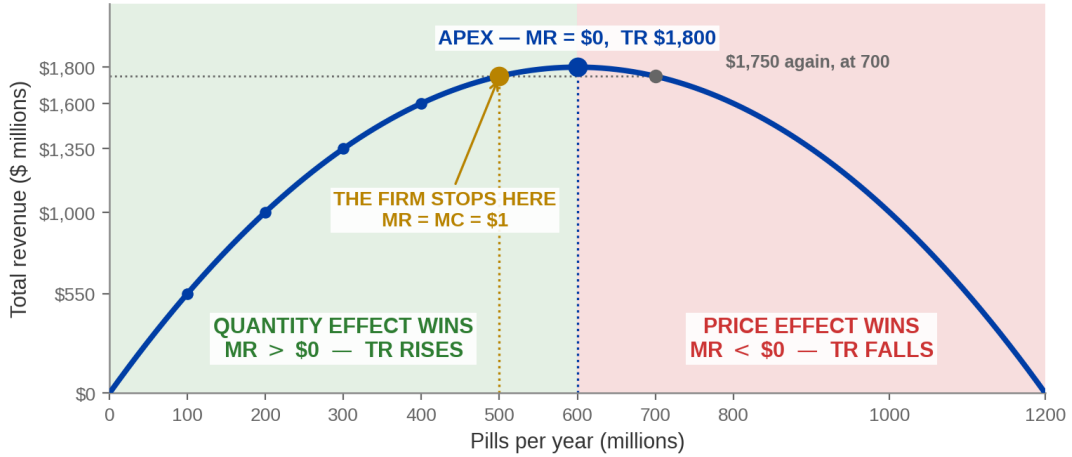


(3.50 − 1.02) × 500 = $1,240 million a year — and it can persist, because the BARRIER TO ENTRY holds.

Cost data from Acemoglu, Laibson & List, Microeconomics 3e, ch. 12

Where Total Revenue Rises, Peaks, and Falls

TOTAL REVENUE = P × Q. It RISES while the QUANTITY EFFECT beats the PRICE EFFECT, PEAKS where MR = $0, and FALLS once the PRICE EFFECT wins.



The monopolist stops at 500 — SHORT of the revenue peak at 600. MAXIMISING REVENUE IS NOT MAXIMISING PROFIT.

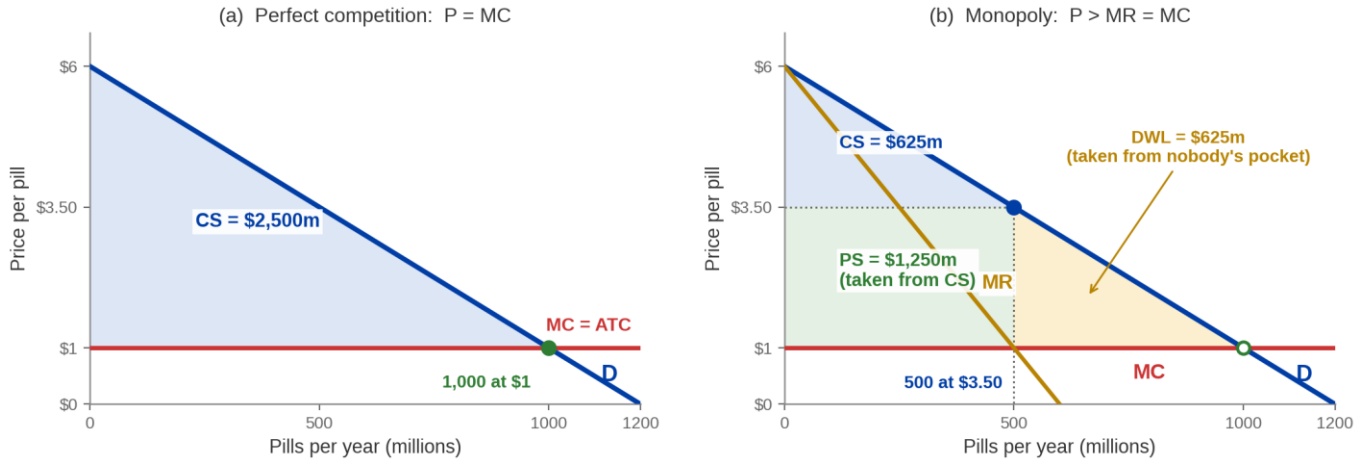
Drawn from the Exhibit 12.5 table on the revenue-and-cost page — Acemoglu, Laibson & List, ch. 12

PART 3

What Monopoly Costs Society

The Same Market, Run Twice

Identical demand, identical costs. Only the number of sellers changes — and SOCIAL SURPLUS falls.



CS falls $2,500m → $625m. $1,250m is TRANSFERRED to the monopolist; $625m is DEADWEIGHT LOSS and nobody receives it.

Market drawn to Acemoglu, Laibson & List, Microeconomics 3e, ch. 12

Competition and Monopoly, Number by Number

One change causes every row: the seller SETS P > MR = MC instead of being handed it.

	Perfect competition	Monopoly	Change
Quantity	1,000m pills	500m pills	Halved
Price	$1.00	$3.50	3.5× higher
Consumer surplus	$2,500m	$625m	Falls $1,875m
Producer surplus	$0	$1,250m	Rises $1,250m
Social surplus	$2,500m	$1,875m	Falls $625m
Deadweight loss	$0	$625m	The whole fall

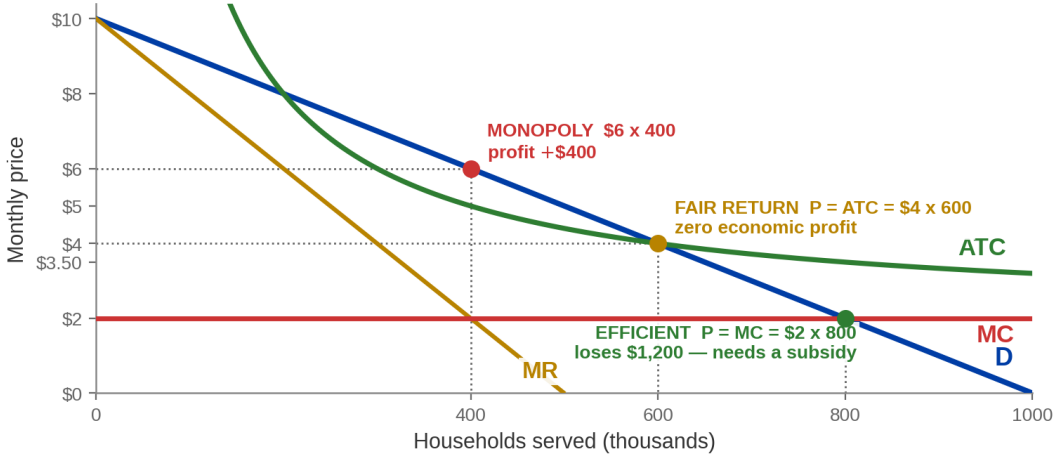
And a MONOPOLIST HAS NO SUPPLY CURVE: a supply curve answers “at price P, how much?” — but a PRICE MAKER is never handed a price.

PART 4

What Policy Can Do About It

Regulating a Natural Monopoly: Two Prices the Law Can Set

P = MC is EFFICIENT but loses money, because ATC lies ABOVE MC. P = ATC is the FAIR-RETURN price: zero economic profit.



Unregulated $6 × 400. FAIR RETURN $4 × 600, zero profit. EFFICIENT $2 × 800 — and a $1,200 loss the taxpayer would have to cover.

Worked example — the natural monopoly of the earlier figure, with demand added

Where Competition Is Possible, Restore It

ANTITRUST POLICY — government action against anti-competitive pricing, restricted output and the deadweight loss they cause.

THE SHERMAN ACT, 1890

Prohibits agreements and actions in RESTRAINT OF TRADE, and monopolising a market. Violations are felonies — fines, and prison sentences.

WHAT A REMEDY LOOKS LIKE



OPEN THE BORDER

A domestic monopolist with no domestic rivals may have plenty of foreign ones. Removing a TRADE BARRIER is competition policy by another name — Chapter 8's argument, applied.

Antitrust attacks the BARRIER TO ENTRY. Price regulation accepts the monopoly and attacks the PRICE. Which one fits depends on whether competition is possible at all.

Chapter 12: The Full Picture

	What it says	Key terms	Where you saw it
One seller	A price maker faces the whole market demand curve, so it must cut price to sell more.	MONOPOLY, MARKET POWER, PRICE MAKER	§12.1 · the two-structure table
Why nobody enters	Legal grant, a key resource, economies of scale, or a network — entry is blocked.	BARRIERS TO ENTRY, NATURAL MONOPOLY	§12.2 · Daraprim, the utility
Marginal revenue	Quantity effect minus price effect, so MR is below price at every quantity.	MARGINAL REVENUE, QUANTITY / PRICE EFFECT	§12.3 · the $5 → $4 cut
The rule	Find Q where MR = MC, go up to demand, read the price. P > MR = MC.	PROFIT MAXIMISATION	§12.4 · 500m pills at $3.50
What it costs	Less output, higher price: surplus transferred to the firm, and a triangle destroyed.	DEADWEIGHT LOSS, TRANSFER	§12.5 · $1,250m moved, $625m lost
What policy does	Attack the barrier with antitrust, or accept the monopoly and regulate the price.	ANTITRUST, FAIR-RETURN PRICE	§12.7 · P = ATC vs P = MC

Not lectured, and in your reading: §12.6, price discrimination — first, second and third degree, and what perfect price discrimination does to consumer surplus.

KEY TAKEAWAYS

Chapter 12: Monopoly

- MARKET POWER — a monopolist is the industry, so it faces the whole downward-sloping demand curve and is a PRICE MAKER.
- BARRIERS TO ENTRY — legal grant, key resource, economies of scale or network effects are what stop entry competing profit away.
- MARGINAL REVENUE is below price, because the QUANTITY EFFECT is offset by a PRICE EFFECT on every unit already being sold.
- THE RULE IS P > MR = MC — choose Q where MR = MC, then read the price off DEMAND, never off MR or MC.
- DEADWEIGHT LOSS — less output at a higher price: some surplus is TRANSFERRED to the firm and some is destroyed, and policy targets one or the other.